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Combustion System Performance and Field Test Results of the MS7001F Gas Turbine

This paper presents the results of the combustion system test of the MS7001F installed at the Virginia Power Chesterfield station. Tests of water and steam injection for NO_x control were performed. Results of emissions, combustor dynamics, and combustor hardware performance are presented. Emissions test results include NO_x, CO, unburned hydrocarbons, VOC, and formaldehyde levels. Combustor dynamic activity over a range of diluent injection ratios, and the performance of an actively cooled transition duct are also discussed. Combustion system mechanical performance is described following the first combustion system inspection.

Introduction

The MS7001F is the first 2300°F (1533 K) heavy-duty gas turbine designed, tested, and commissioned. This step-change in firing temperature for the new machine presented design challenges for most of the gas turbine components including the combustion system (Brandt, 1988). Component testing and development of the combustion system were accomplished, in large part, on single combustor test stands at GE's turbine test laboratories. The first 7F machine was no-load and load tested at the Greenville, SC, test facility in 1988 and 1989, shipped and installed at its Chesterfield, VA, site (Virginia Power), and instrumented and tested during the spring of 1990. The amount of instrumentation and the detail embodied in the test plan are greater than in any previous machine test. The success of this field test is due to the attention to detail and the extensive component testing that occurred prior to machine assembly. The results obtained confirm the quality of the system design, and allow for quick response to changing market needs such as upgraded output or improved emissions guarantees.

System Description

The combustion system for the GE MS7001F gas turbine is illustrated in Fig. 1, which shows one of the 14 combustion chambers of the machine in cross section. The use of multiple, reverse flow combustion chambers is typical of the GE heavy-duty, industrial gas turbine design. The 13 deg cant of the combustors allows for a more compact combustor and gas turbine design. The combustion system consists of a fuel nozzle assembly, a flow sleeve, a cap and liner assembly, and a transition piece. The combustion system incorporates advanced cooling technology to permit firing temperatures of 2350°F (1561 K). Ignition is achieved with a spark plug in 2 of the 14 chambers with cross firing connections to ignite the balance of the chambers. Successful ignition is sensed with four ultraviolet flame detectors located nearly opposite to the location of the spark plugs.

The air flow paths for the combustion system are also shown in Fig. 1. After leaving the diffuser area of the compressor, air flows past the transition pieces and forward toward the head end of the liner. The air splits between two parallel paths before entering the liner. Some of the air flows through the holes provided in the impingement sleeve of the transition piece for cooling; the balance of the air passes through an array of multiple holes in the flow sleeve. The air, now outside the liner, experiences dilution, mixing, cooling, and enters holes in the liner sleeve and passes through the cap of the liner, where the air and fuel mix. Inside the liner, the air and fuel react and flow downstream through the transition piece before entering the turbine section.

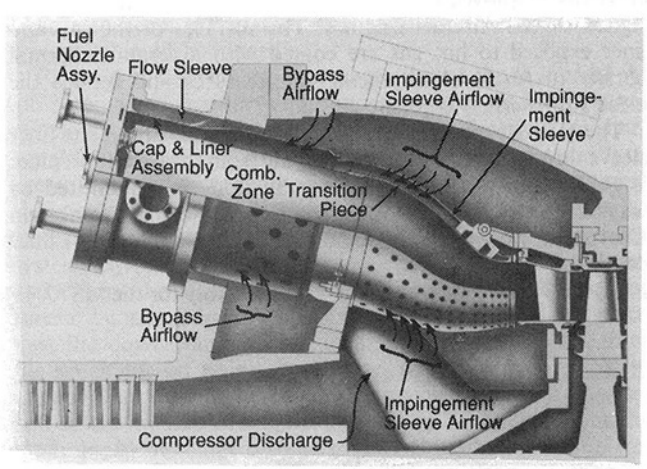

Fig. 1 MS7001F combustion system

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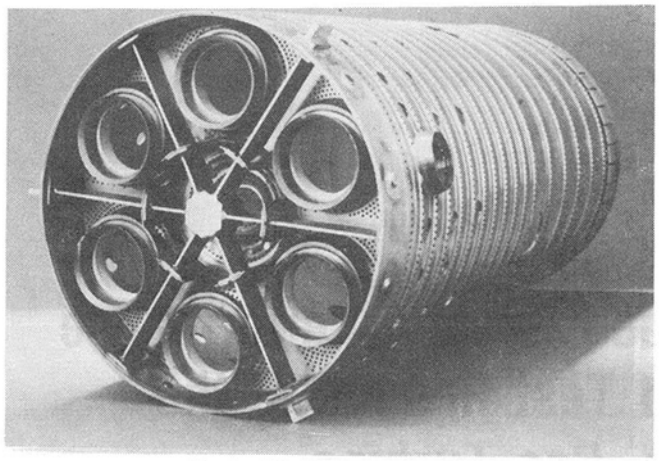

Fig. 2 MS7001F cap and liner

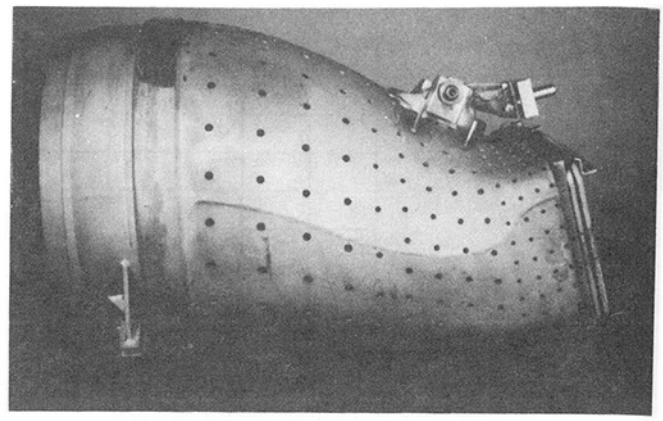

Fig. 3 MS7001F transition piece assembly

The fuel nozzle assembly is designed to allow use of either gas or distillate fuel and to transfer from one fuel to the other during operation. Six individual, dual-fuel nozzles are attached to an internally manifolded cover. Each fuel nozzle consists of a swirl tip, an atomizing air tip, and a distillate spray nozzle. Gas fuel is injected through metering holes in the swirler. Continuous atomizing air is used with the distillate fuel spray. Water or steam for NO_x control also enter through the combustion cover. The multinozzle design offers two significant advantages over single-nozzle designs. First, it allows more thorough mixing and control of the fuel and air in the reaction zone, resulting in a shorter flame length and improved combustor performance. Second, it produces significantly lower noise levels than single-nozzle combustors and thereby improves combustor component wear. The low noise benefits of the multinozzle configuration were first realized on the GE MS7001E multinozzle quiet combustor (EPRI, 1985; Fitts et al., 1990).

The cap and liner assembly, shown in Fig. 2, consists of a slot film cooled, 14 in. (0.35 m) diameter, 30 in. (0.76 m) long liner sleeve and a multinozzle cap assembly. The slot cooling design employed has evolved from earlier applications on other GE gas turbine liners. The relatively short liner length is another benefit of the multinozzle design, which produces a shorter reaction zone than single-nozzle combustors. The shorter liner has less surface area to cool. Thus, in spite of the higher firing temperature, the MS7001F liner operates at temperatures less than or equal to those of earlier designs. The cap is film cooled in the center and impingement cooled in the region surrounding the fuel nozzles. The impingement/splash plate design employed on the MS7001F cap is similar to that currently employed on the domes of annular combustors designed by GE Aircraft Engines. The surfaces of the cap and liner exposed to hot gas are coated with a ceramic thermal barrier to further limit metal temperatures and reduce the effects of thermal gradients.

The flow sleeve forms an annular air space about the liner sleeve through which spent cooling air from the transition piece is guided and the balance of combustion air is admitted by way of the array of holes. The flow sleeve also acts to maintain a sufficient flow velocity on the back side of the liner for liner cooling.

Figure 3 shows the transition piece assembly for the MS7001F gas turbine. The transition piece assembly employs an innovative cooling design, which has allowed the increases in compression ratio and firing temperature that are required in an advanced gas turbine.

The transition piece directs the hot gases from the exit of the liner sleeve to the inlet of the turbine. Such a transition duct is typical of GE gas turbines and has been successfully employed on thousands of gas turbines in various configurations. All earlier designs, however, are convectively cooled by the passive flow of compressor discharge air around the transition piece. Good heat transfer is effected on the inner radial and side surfaces of the transition piece by virtue of the high-velocity air flow and correspondingly high heat transfer coefficients over these surfaces. However, the outer radial panel, particularly toward the aft end of the transition piece, is in a shadow where the air is relatively stagnant and the heat transfer coefficient is low. As a result, the outer radial panel toward the aft end of the transition piece experiences the highest temperatures which are thus design limiting.

In contrast, the MS7001F transition piece is surrounded by a sleeve in which an array of holes is provided. Air flows through the holes and impinges on the back side of the transition piece, thereby providing an effective impingement-type cooling of the surface. The shape of this "impingement" sleeve roughly parallels that of the transition piece with the gap between the two increasing from the aft end to the forward end to accommodate the increase in air flow from each of the consecutive rows of cooling holes. The use of impingement cooling allows more precise control of the cooling of the transition piece while having only a small, adverse effect on the combustion system's overall pressure drop.

The cooling of the transition piece provided by the impingement sleeve results in significantly more uniform metal temperatures on the transition piece, thereby lowering thermal stresses and allowing higher hot gas and compressor discharge temperatures to be accommodated while maintaining metal temperatures consistent with GE experience. The internal surfaces of the transition piece, those exposed to the hot gas, have a ceramic thermal barrier coating applied to limit the maximum metal temperature further.

Test Plan and Objectives

The test plan for the field test of the MS7001F was quite

Nomenclature

IGV = inlet guide vane
ppbv = parts per billion by volume
ppmvd = parts per million by volume, dry
ppmvw = parts per million by volume, wet
TF = firing temperature

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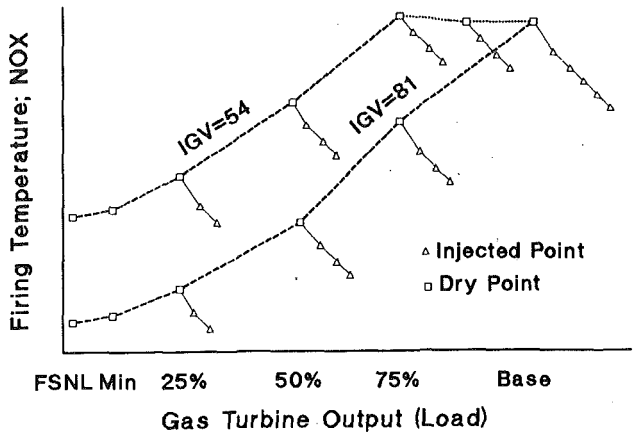

Fig. 4 MS7001F field test emissions plan

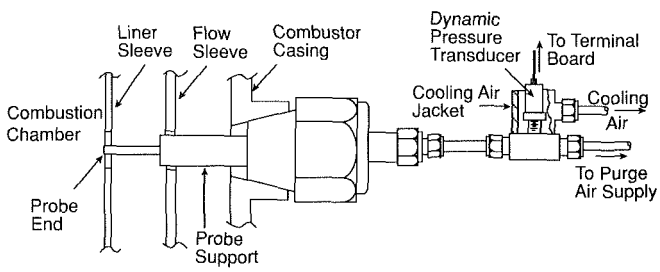

Fig. 5 Schematic arrangement of combustion chamber dynamic pressure probe

comprehensive as described briefly by Cincotta (1990). A large portion of this test plan was specifically for measurement and evaluation of the performance of the combustion system. Previous testing at the Greenville factory test had evaluated ignition capability and part-load performance as well as full temperature performance (albeit at reduced air flow due to the load absorbing limits at the factory test site). The overall field tests involved synchronization on oil and gas, full load, mechanical design, various configurations of water and steam injection, aeroperformance, modulated inlet guide vanes (IGV), and operating performance. The combustion emissions test plan was detailed enough that nearly all combustion requirements were investigated as part of the emissions test plan. Figure 4 presents the emissions test points in schematic fashion. For each fuel, gas or oil, points were run at various combinations of firing temperature and inlet guide van angle, which corresponded to load points of approximately 0, 25, 50, 75, and 100 percent of base load. On most of these test points diluent injection was also tested; these points are shown as branches off the main lines of constant IGV angle.

The noninjected points were run to give emissions values for the dry case. The injected points were run to determine required diluent flow for NO_x control and the performance of the combustion system, injection control systems, and the 7F machine itself. Firing temperatures above base load and guide vane angle greater than 81 deg were also explored. Diluent/fuel ratios were selected to straddle the ratios required for NSPS (New Source Performance Standard), and 42/65 (42 ppm @ 15 percent O_2 with natural gas, 65 ppm @ 15 percent O_2 with distillate). NSPS and 42/65 were required to check the lab data for NO_x abatement with water and steam and to provide data for the control curves for the customer whose permit required operation at 42/65. In addition to the emissions measurements, the fully array of combustion data was taken for each test point. Water injection and steam injection were both evaluated. A special water injection skid and associated hardware and piping were installed for the water injection points; afterwards the machine was configured and tested in the steam injection mode which the customer had purchased. At the completion of the testing and prior to commissioning the machine, a complete combustion system inspection was performed to ensure GE and the customer of the integrity of the hardware.

Table 1 Combustion system instrumentation

	MEASURED QUANTITIES OR SENSORS (TOTAL SENSOR COUNT = 524)						
COMPONENT	METAL TEMP.	DYNAMIC PRESS.	DIFF'L PRESS.	STATIC PRESS.	HOT GAS TEMP.	HOT GAS PRESS.	STRAIN GAGES
FUEL NOZZLE ASSEMBLY	20	7	7				
FLOW SLEEVE ASSEMBLY				36			
CAP & LINER ASSEMBLY	92	14					20
TRANSITION PC. ASSEMBLY	136			116	48	8	20
TOTALS	248	21	7	152	48	8	40

Instrumentation Description

Four combustion chambers were instrumented, in various combinations, with thermocouples to measure metal temperatures on the liner and transition piece, thermocouples to measure combustor exit profile, pressure taps to monitor air flow distribution, and strain gages to quantify the dynamic response of the liner and transition piece. In addition, all chambers were instrumented with dynamic pressure transducers to measure combustion chamber pressure oscillations and static pressure taps to assess uniformity of air flow among the chambers. In total, the combustion system was instrumented with more than 520 sensors. Table 1 illustrates the distribution of the various sensors among the combustor's component parts. The data collected by these sensors were in addition to the data typically captured by the gas turbine's control system, such as fuel flow, water or steam injection flow, exhaust temperatures, etc.

Dynamic pressure oscillations within the combustion chamber are measured with a radial probe that is approximately flush with the inside wall surface of the liner sleeve. The probe penetrates the outer casing of the unit's pressure vessel. Once outside the machine, a dynamic pressure transducer is affixed to the probe. Figure 5 is a schematic illustration of the dynamic pressure probe. Each of the gas turbine's 14 combustors was outfitted with a dynamic pressure probe to ensure that worst chamber behavior could be monitored. Dynamic pressures in each of the fuel, diluent, and atomizing air supply manifolds were also recorded so that any coupling to the combustor dynamics could be identified.

Figure 6 is a schematic representation of the layout of the gas turbine and auxiliary equipment as well as the location and proximity of the control and data rooms. Steady-state data, consisting of gas and metal temperatures and static and total pressures, were recorded periodically on a digital data acquisition system. Control system data relevant to the combustion system were also accessed and recorded by the data acquisition system. Dynamic data from piezo-electric pressure transducers and electrical resistance strain gages were recorded in analog format on magnetic tape. Data were recorded during all start-ups, shutdowns, and loading transients, as well as at each of the specified test points once steady-state operation was attained. Data were gathered for dry operation as well as water and steam injection for both gas and distillate fuels over the load range of the gas turbine to characterize the behavior of the combustor.

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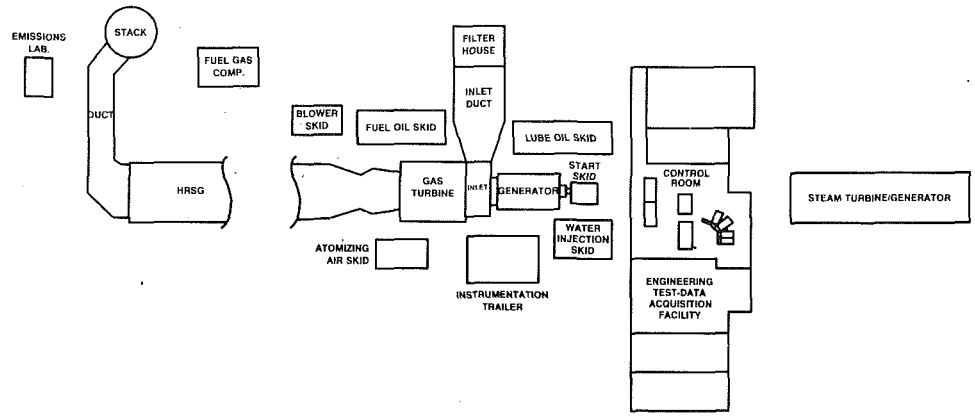

Fig. 6 Plant layout

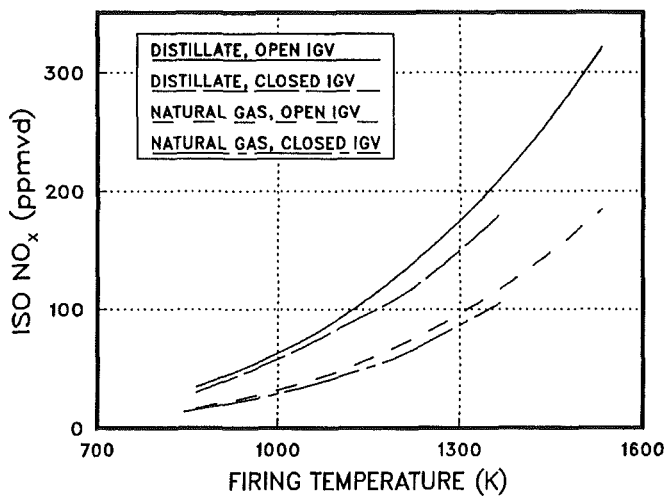

Fig. 7 NO_x at ISO conditions as a function of firing temperature

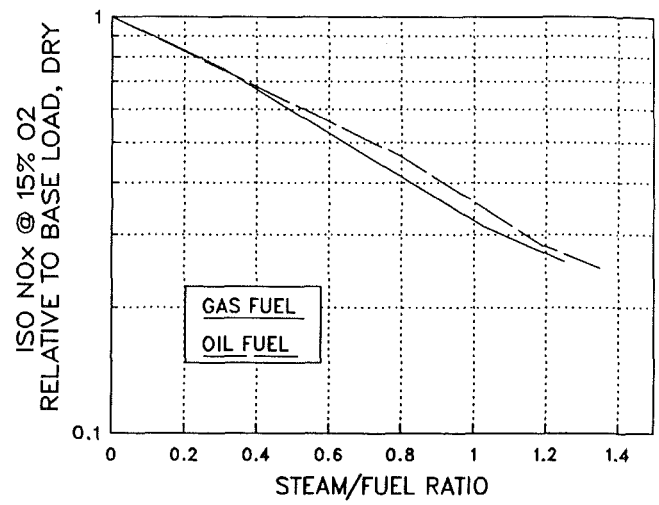

Fig. 8 Effect of water injection on NO_x emissions

Test Results

Emissions. Throughout the testing of the MS7001F, GE had a contracted vendor on site measuring exhaust emissions. The exhaust was sampled for oxides of nitrogen (NO_x), carbon monoxide (CO), unburned hydrocarbons (UHC's), volatile organic compounds (VOC's), aldehydes, particulates, and nitrous oxide (N_2O). In addition visible emissions were recorded. Emissions were measured without diluent injection and with water and steam for NO_x control.

Oxygen Traverse. Before emissions sampling began, a duct traverse was done to measure oxygen. The traverse was done in the ducting between the HRSG and the stack. As expected, the O_2 variation was very small and the balance of the testing was carried out using only a single point sample. Samples for particulates, aldehydes, etc., were also single point samples but were taken at a different point.

NO_x. NO_x results, without steam or water injection, are shown in Fig. 7 corrected to ISO humidity, temperature, and pressure. The pressure corrections are based on ambient pressures rather than compressor discharge pressures as suggested by EPA. This figure shows NO_x for minimum and maximum IGV angle for both gas and distillate fuels.

The machine was tested with both steam and water for NO_x abatement. The base load levels of NO_x corrected to ISO conditions and 15 percent oxygen relative to base load, without water or steam injection, are shown in Figs. 8 and 9, respectively, for both fuels. From these figures, approximate diluent/fuel ratios at base load required for NO_x reduction to various

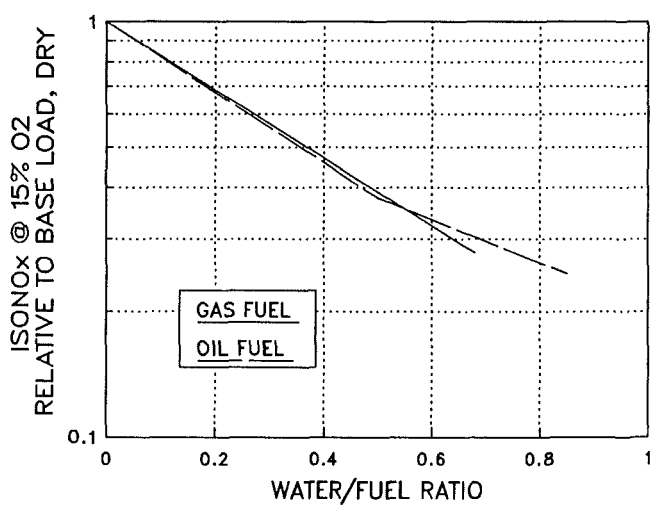

Fig. 9 Effect of steam injection on NO_x emissions

NO_x control codes can be determined (these are given in Tables 2 and 3).

NO_2. In addition to total NO_x, NO_2 was also measured at each data point. Normally, the emissions sample is run through a converter, which changes all NO_x to NO. Thus, total NO_x is measured as NO. To measure NO_2, the sample bypassed the converter and was fed directly into a NO_x meter, measuring

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Table 2 Water/fuel ratios required for various NO_x emissions levels

FUEL	NOx level	W/F
Natural Gas	NSPS	.16
Natural Gas	42 ppmvd	.68
Distillate	NSPS	.47
Distillate	65 ppmvd	.85

Table 3 Steam/fuel ratios required for various NO_x emission levels

FUEL	NOx level	S/F
Natural Gas	NSPS	.40
Natural Gas	42 ppmvd	1.25
Distillate	NSPS	.85
Distillate	65 ppmvd	1.35

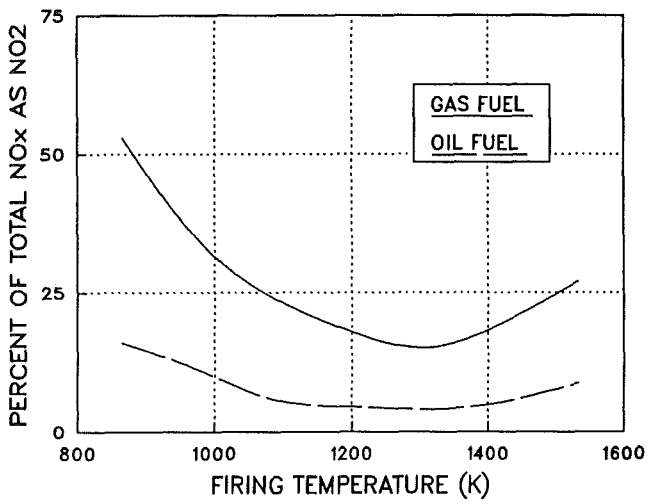

Fig. 10 NO_2 emissions as a function of firing temperature

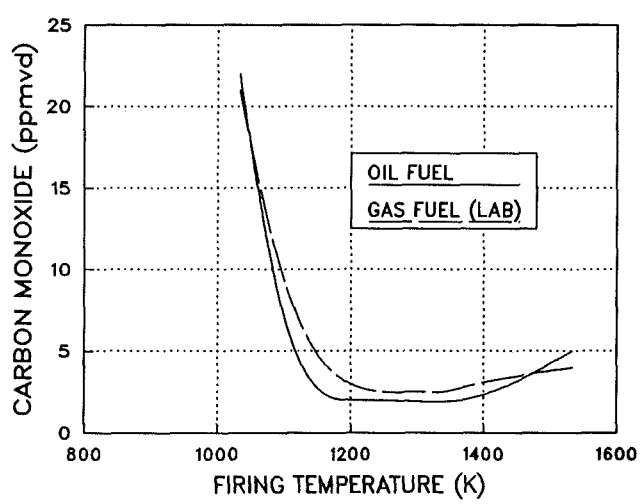

Fig. 11 Carbon monoxide emissions as a function of firing temperature

actual NO in the exhaust. By subtracting this number from the total NO_x number, the concentration of NO_2 is obtained. Figure 10 summarizes the results, showing percent of total NO_x as NO_2 for both fuels at 81 deg IGV.

Note that for gas fuel, the percentage of total NO_x present as NO_2 is higher than for distillate. This is probably due to high unburned hydrocarbon levels present during all natural gas testing. During natural gas testing, a valve between the gas manifold purge and gas supply was not functioning properly.

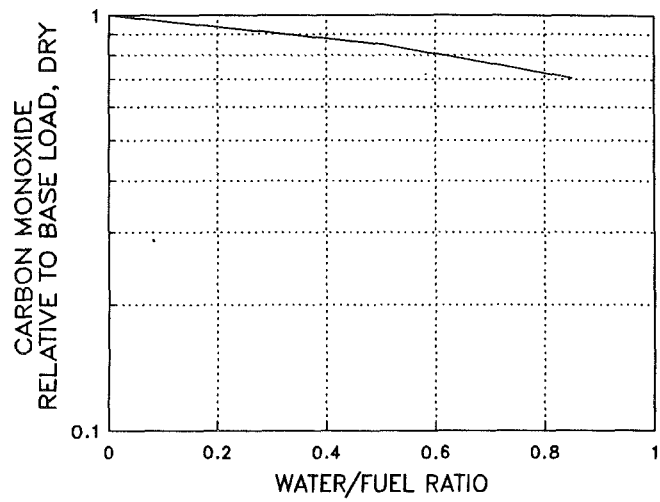

Fig. 12 Effect of water injection on carbon monoxide emissions

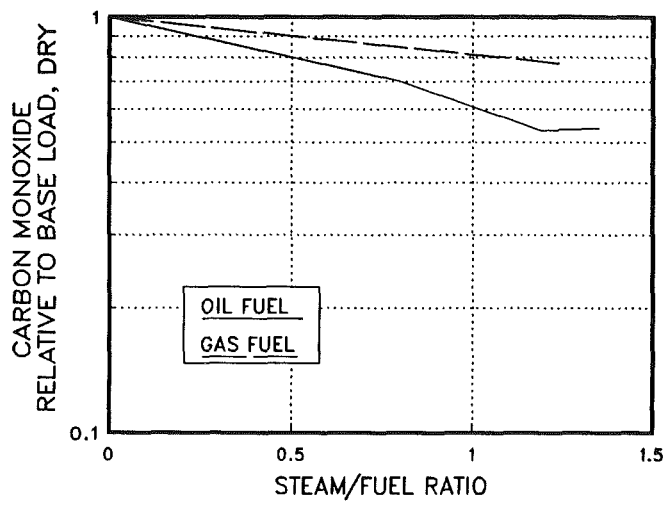

Fig. 13 Effect of steam injection on carbon monoxide emissions

As a result, some natural gas was vented to the atmosphere near the inlet filters of the gas turbine. Some of this gas was ingested into the turbine and bypassed the combustion system. This gas showed up as unburned hydrocarbons in the exhaust. Evidence exists to suggest that high levels of unburned hydrocarbons accelerate the conversion of NO to NO_2, with little or no effect on total NO_x (Hori, 1988).

Carbon Monoxide. Carbon monoxide emissions as a function of firing temperature without water or steam injection are presented in Fig. 11 for distillate oil and natural gas. The gas data are based on laboratory observations. Due to the leaking gas manifold purge valve, CO data gathered while burning natural gas are not accurate. Based on the agreement between laboratory and field CO data gathered on distillate fuel, the CO curve for gas from the laboratory data is used here.

The effect of water and steam injection on CO emissions at base load is shown in Figs. 12 and 13. Because of the leak in the purge valve, CO data on gas fuel are limited to those gathered during the compliance test that was performed after the engineering testing and after the leak was corrected. The base load compliance test data are limited to the single point at a steam fuel ratio of 1.25 shown in Fig. 13. Since the machine was only compliance tested in the steam injection configuration, no water-injected data are shown for gas fuel.

Unburned Hydrocarbons. Figure 14 shows the total un-

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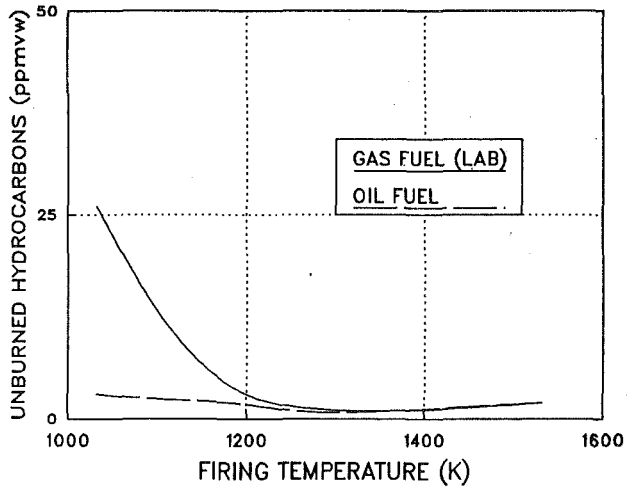

Fig. 14 Unburned hydrocarbon emissions as a function of firing temperature

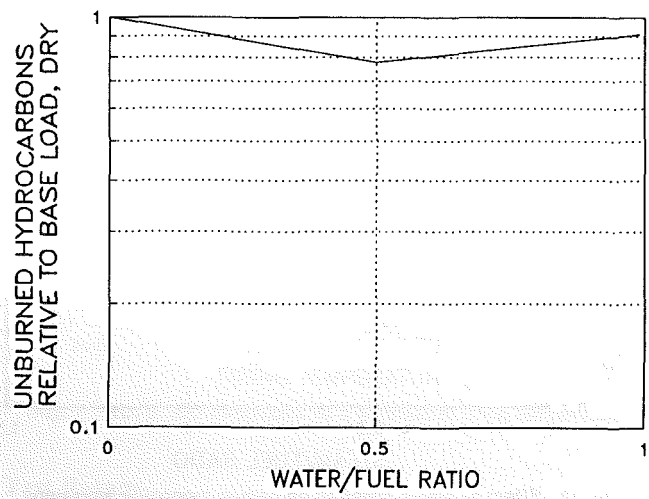

Fig. 15 Effect of water injection on unburned hydrocarbon emissions

burned hydrocarbon emissions for distillate and gas fuels with no diluent injection as a function of firing temperature. As with the CO data, the gas-fired data shown in the figure are based on laboratory results.

The effect of water injection on total unburned hydrocarbons at base load is shown in Fig. 15. Once again, because of the unreliable natural gas data, no gas curve is shown for water injection. No figure is shown for steam injection because unburned hydrocarbon emissions showed no change with steam injection for either fuel at base load.

VOC's. Volatile Organic Compounds have been defined as all unburned hydrocarbons excluding methane and ethane. The levels measured were all in the range of one ppm or less and therefore are not shown in figure form.

Visible Emissions. Visible emissions were measured by a certified visible emissions observer (VEO). Opacity as determined by the VEO was measured on distillate fuel at base load both with no water injection and with a water/fuel ratio of about 1.0 and again at no load on distillate. No VEO readings were taken while burning natural gas; however, the worst case for smoke usually occurs when burning distillate. All opacity readings taken for these points were either zero or 5 percent.

Particulates. One particulate sample was taken for each fuel at base load. For distillate the sample was taken with a water/fuel injection ratio of about 1.0. The gas sample was taken with steam injection at a steam/fuel of about 1.1. For both samples the particulate emissions rate was less than the guaranteed values of 7 and 14 pounds per hour for the 7F on gas and distillate, respectively.

Table 4 Summary of aldehyde emissions (all values ppbv)

FUEL	DIST	DIST	DIST	GAS	GAS	GAS
LOAD (MW)	0.0	140.5	150	10.7	14.5	154
w/f	0.0	0.0	0.99	N/A	N/A	N/A
s/f	N/A	N/A	N/A	0.0	0.0	1.07
Acetaldehyde	1270	37.8	21.3	47	15.5	17.4
Acetone	308	210	144	96.8	723	723
Acrolein	530	<3.30	<2.42	<2.16	<2.56	<2.38
Benzaldehyde	89.7	<1.75	<1.28	<1.14	<1.35	<1.25
Butyraldehyde	123	<2.57	<1.88	<1.68	<1.99	<1.85
Formaldehyde	6970	946	485	10800	93.6	1130
Hexaldehyde	132	<1.85	<1.35	<1.21	<1.43	<13.1
Isovaleraldehyde	18	61.0	38.3	1.55	<1.84	<13.1
Veraldehyde	309	24.0	18.2	12.7	1.84	503

Aldehydes. Six exhaust samples were collected in an impinger train to check for aldehyde emissions; three for distillate and three for natural gas fuel. The impingers contained acidified 2,4-dinitrophenylhydrazine. Probe and impingers were recovered with methylene chloride. Each sample was analyzed with high-performance liquid chromatography for nine different aldehydes. The results appear in Table 4. Formaldehyde was the most prevalent specie detected with a maximum value of about 7 ppmv on distillate and 11 ppmv on gas. All other readings were on the order of 1 ppmv or less. The high readings on gas fuel may have been effected by the leaking gas purge valve. The highest reading on distillate occurred at minimum load with no water injection. The oil nozzles at this point were configured to runs with a mix of oil and water and as such were sized larger than conventional oil nozzles. The low fuel flow through this nozzle having a higher flow number probably resulted in poor atomization of the fuel. This would be supported by the higher hydrocarbon readings: 82 ppmvd versus 21 ppmvd for a similar minimum load point with conventional oil nozzles.

N_2O. For 11 of the test points, samples of the exhaust were drawn into tedlar bags and sampled for nitrous oxide (N_2O). The results appear in Table 5. For all of the points no N_2O was detected with the detection limit set at 500 ppb for some points and 260 ppb for others.

Combustor Dynamic Pressures. Figure 16 illustrates typical frequency spectra of the MS7001F combustor at base load, dry, and at a water-to-fuel ratio of 1.0. In both cases the dynamic activity is broad band with no significant discrete peaks. The effect of water injection is seen here to shift the entire spectrum to a higher overall level. At even higher injection rates, there is an enhanced response around the combustor's lowest longitudinal mode. This enhancement, however, is still fairly broad in nature. A distributed noise signature provides a minimal stimulus to the combustion system components. GE's experience with such a low and broad band signature shows a strong relationship between such dynamic behavior and the wear behavior of the combustor. That is, with low dynamic levels, the wear of the combustion system is correspondingly low. This is important as the combustor is subjected to increasing levels of diluent injection for NO_x control or power augmentation.

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Table 5 Summary of N_2O emissions

FUEL	LOAD (MW)	W/F	S/F	N_2O (ppbv)
OIL	0.0	0.0	N/A	ND*
OIL	9.6	0.0	N/A	ND*
OIL	10.6	0.0	N/A	ND*
OIL	140.5	0.0	N/A	ND*
OIL	147.5	0.0	N/A	ND*
OIL	150.0	0.99	N/A	ND*
OIL	150.0	1.42	N/A	ND+
GAS	10.7	N/A	0.0	ND+
GAS	12.8	0.0	N/A	ND+
GAS	144.5	N/A	0.0	ND+
GAS	155.0	N/A	1.07	ND+

* not detected at <500 ppb
\+ not detected at <260 ppb

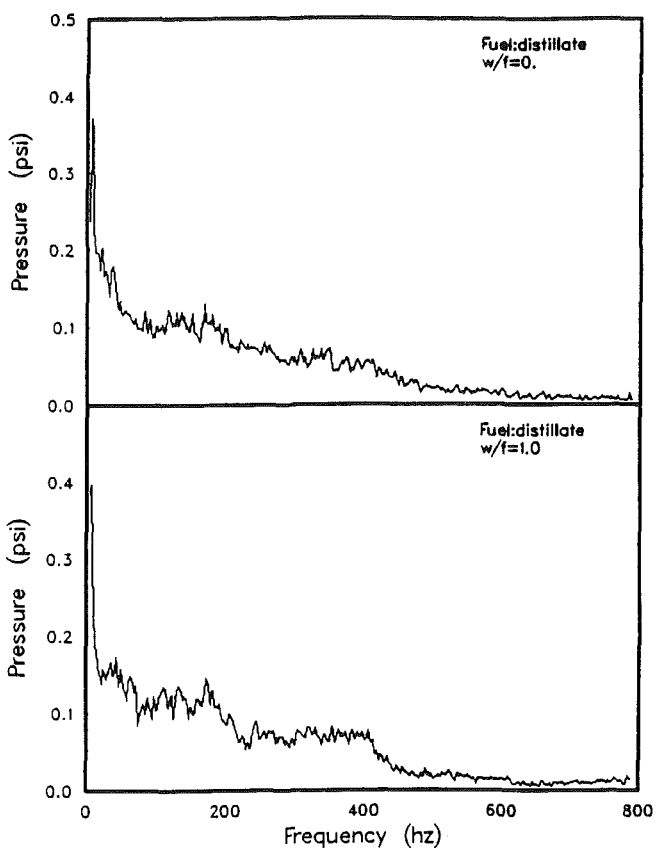

Fig. 16 Combustion dynamic pressure spectra

The remainder of the combustor dynamic pressure data is presented as overall (rms) levels of the signal from 0–800 Hz. The 0–800 Hz range captures all relevant combustion acoustic information, which generally drops off quickly above 400–500 Hz in these large combustors. For reference, the lowest acoustic resonant mode of this combustor is a longitudinal standing wave, which varies in frequency from about 200 Hz at zero load to about 300 Hz at base load. All combustor dynamic pressure plots have been normalized to base load, dry on distillate fuel.

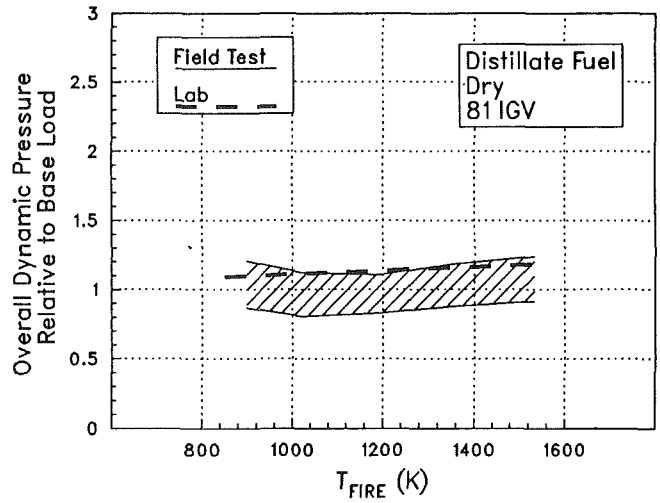

Fig. 17 Combustor dynamic pressure as a function of firing temperature for distillate fuel

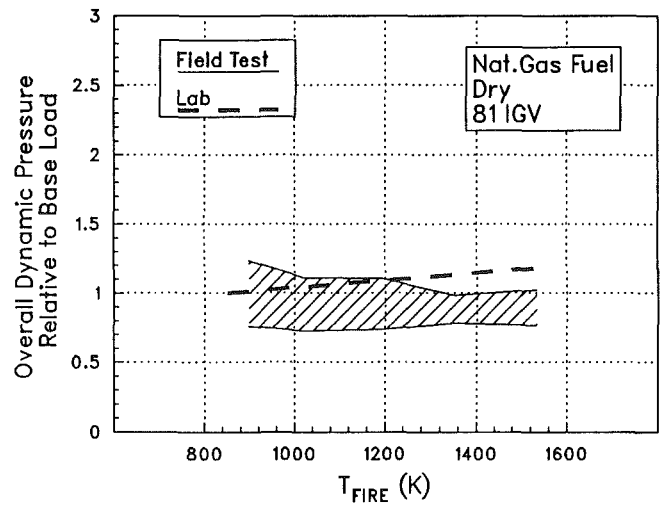

Fig. 18 Combustor dynamic pressure as a function of firing temperature for gas fuel

Figure 17 is a plot of the overall combustor dynamic pressure as a function of firing temperature on distillate fuel dry. The curve is an average of all 14 chambers while the shaded region shows the maximum to minimum range. All shown are the dynamic pressure levels seen during laboratory single chamber tests. As seen, the laboratory data fall near the high end of what is seen in the machine. The relative flatness of the curve as load increases is a measure of the inherent quietness of the multinozzle combustor design. Figure 18 illustrates the same data but for gas fuel operation. The same comments regarding the dynamic pressure activity apply to gas fuel as for distillate fuel.

To illustrate the effect of diluent injection on the dynamic pressure oscillations, the dynamic pressure is plotted as a function of the diluent to fuel ratio. Figures 19–22 show these plots for both fuels, for both diluents and for both base load (100 percent) and part load (50 percent). The field test data are compared to laboratory data in each figure. In addition, each figure contains lines corresponding to the diluent to fuel ratios required for various levels of NO_x control. The increase attributable to water injection can be determined by referring to the point of zero water to fuel ratio (dry).

The results for distillate fuel show that as diluent injection is increased, the dynamic pressure level increases only slightly.

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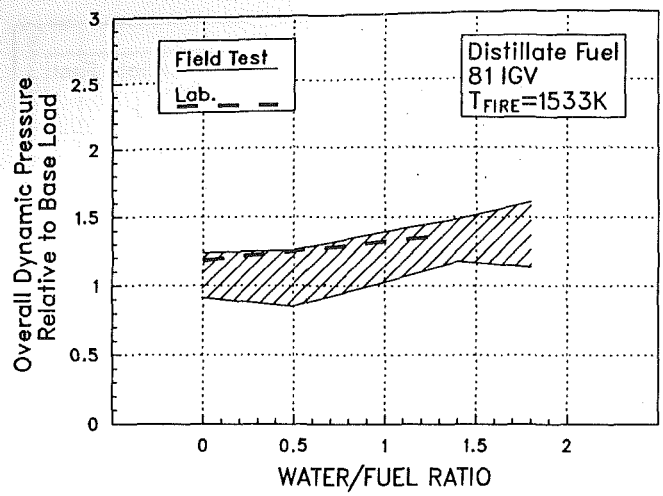

Fig. 19 Effect of water injection on combustor dynamic pressure for distillate fuel

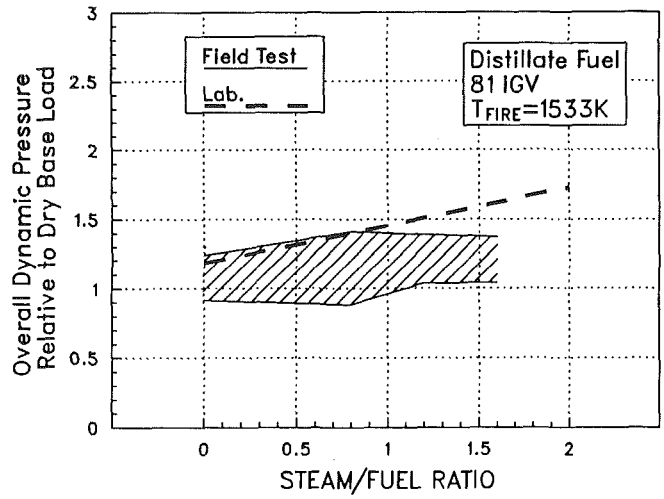

Fig. 21 Effect of steam injection on combustor dynamic pressure for distillate fuel

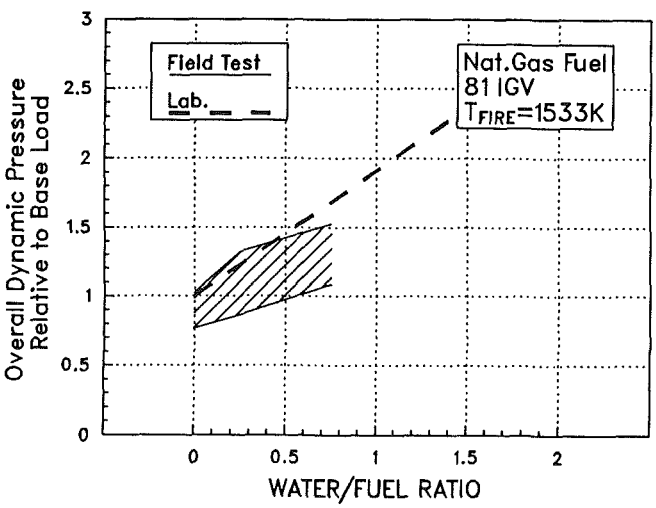

Fig. 20 Effect of water injection on combustor dynamic pressure for gas fuel

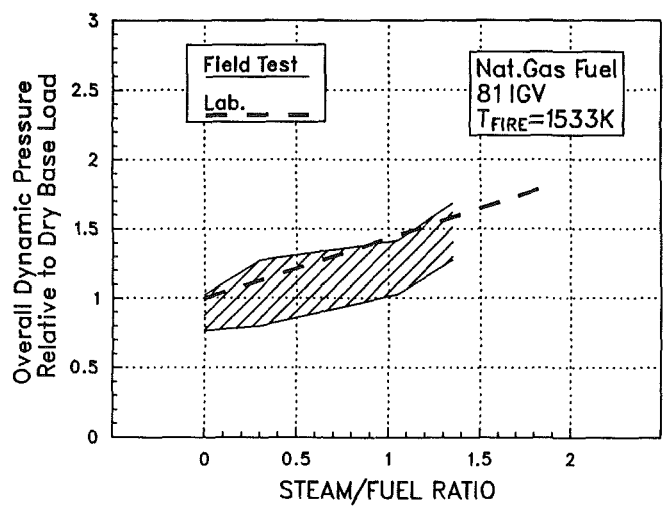

Fig. 22 Effect of steam injection on combustor dynamic pressure for gas fuel

With water injection, the increase is about 0.25 psi or about 20 percent as the water-to-fuel ratio approaches 2:1. With steam injection, an increase of about 15 percent is seen at a steam/fuel ratio of about 1.6.

On gas fuel, the increase in dynamic pressure levels with increasing diluent injection is more pronounced than that seen for distillate fuel. This is probably attributable to the interaction of the gas fuel and water flows, which are both injected into the reaction zone as discrete jets in close proximity to one another. Although this injection scheme produces higher dynamic levels, it does yield good emissions performance. These levels are still well within the range of experience of other GE combustors.

Overall, the dynamic pressure behavior of the GE MS7001F gas turbine combustor was very good. The level of dynamic pressure oscillations was at the low end of GE's experience. The quiet behavior should prove to be beneficial in promoting hardware longevity. The low levels are an inherent result of the combustor's design features, which were developed specifically to reduce the level of dynamic pressure activity.

Combustor Dynamic Strains. The liner and transition piece were instrumented with several dynamic strain transducers. These gages were located at points of high stress such as near the hardware support points or near stress concentrators such as liner holes. The measured strain levels were quite low and subject to a poor signal-to-noise ratio; therefore, a detailed quantitative assessment of the strain data was not made. The maximum peak-to-peak dynamic strains observed were about 50 μin./in. and occurred during base load operation on the liner.

Combustor Metal Temperatures. The combustor metal temperatures for base load, dry operation are shown in Figs. 23 and 24. Figure 23 shows the axial distribution of liner metal temperature. The liner thermocouples were installed on the back side of the liner wall. There is one thermocouple for each of the 16 film-cooled panels located at the point on each panel where the highest metal temperatures are expected. The temperatures are uniform over the first 80 percent of liner length but rise about 200°F (110 K) in the last 20 percent. Because of the hotter tail end, the liner material changes from Hastelloy-X to Haynes-188 over the last 35 percent of the liner. The Haynes-188 alloy has 100–200°F (55–110 K) better temperature capability than the Hastelloy-X.

Figure 24 compares the circumferential temperature distributions typical of the MS7001F impingement-cooled transition piece with those of a passively cooled transition piece from an

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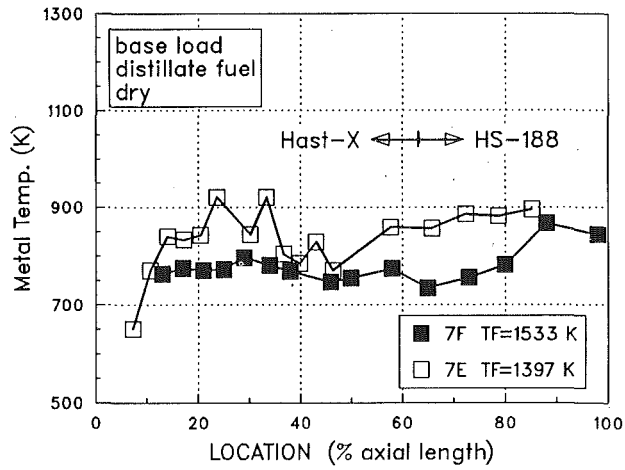

Fig. 23 Liner metal temperature distribution

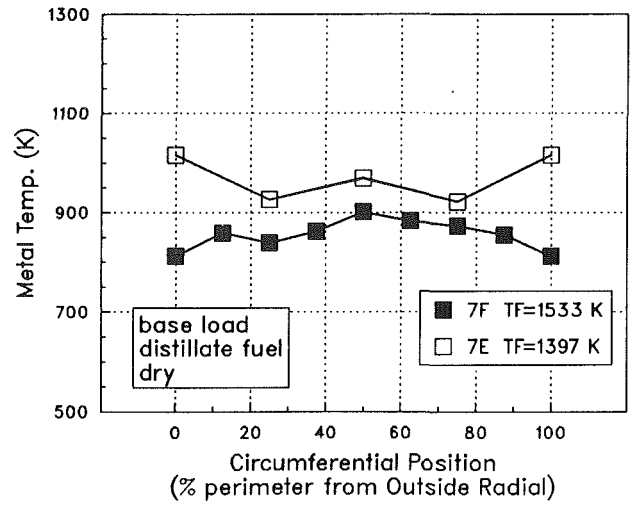

Fig. 24 Transition piece circumferential temperature distribution

MS7001E machine. The passively cooled, MS7001E transition piece shows a decided temperature rise on the outer radial panel, which coincides with the location of the mounting features. Higher stresses in proximity to the mounting feature in combination with temperature level therefore limit the capability of the transition piece when the bulk of the part has not approached the maximum operating temperature limit. The side walls of the MS7001F design show a moderate rise in metal temperature; however, as the stress levels in the side wall are relatively low, the metal temperature can be allowed to rise without creating a concern. Not only is the temperature more uniform as a result of the impingement cooling, but overall the temperatures are lower for the MS7001F design. Therefore, as a direct result of the impingement-cooled design and the consequent uniformity of temperature, the transition piece should exhibit improved longevity from enhanced low cycle fatigue and creep behavior.

The circumferentially averaged axial temperature distribution on the transition piece is shown in Fig. 25. Each temperature in the figure is the average of eight thermocouples located around the circumference of the transition piece. The thermocouples were positioned so as not to be directly in line with the impingement cooling jets, to ensure a temperature measurement that was not overly optimistic. The temperature distribution is seen to be fairly flat and is below the imposed temperature limit for the transition piece material.

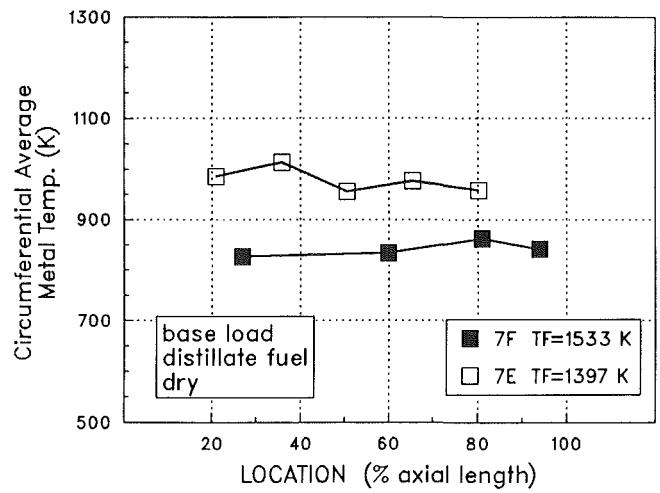

Fig. 25 Transition piece axial temperature distribution

Inspection Results

The combustion system inspection performed at the completion of the engineering test showed no operationally related hardware distress. One liner was inadvertently damaged during disassembly but was readily repaired. The combustion system was reassembled and the unit commissioned. During the engineering test, the unit accumulated 209 fired hours and 37 fired starts. It was recommended that the next inspection be performed after 3000 hours of operation.

At the first combustion inspection after commercial operation of the unit, an additional 3,965 fired hours and 96 fired starts had accumulated. The unit had operated primarily on gas fuel. The unit had operated for approximately six hours on distillate fuel prior to the planned shutdown for the inspection. The unit was brought down on a fired shutdown with distillate fuel.

The overall condition of the combustion system was very good. The inspection results indicate that the hardware is capable of at least 8000 hour inspection intervals. While some unexpected conditions were noted, they will not adversely affect the 8000 hour inspection interval.

Approximately half of the gas swirl tips were cracked to the point that reuse was not recommended. Since the inspection, a failure analysis and flow study were completed to understand better the cause of the swirl tip cracking. The work showed that the swirl tip is subject to large thermal gradients, causing thermomechanical fatigue. A modification that significantly reduces the magnitude of the stresses was developed and will be installed at the next scheduled outage.

The fuel nozzles also exhibited a buildup of hard carbon on the face. The carbon buildup only occurs with distillate fuel operation and is worst at low loads and during a fired shutdown. Since the inspection, flow studies and laboratory tests have identified design changes that will significantly reduce, if not eliminate, the buildup. Improved design fuel nozzles will be installed at the next scheduled outage.

Two liners exhibited a small perforation, approximately 0.5 in.^{2} (3.2 cm^2) of the splash plate and impingement plate. The carbon buildup on the gas tip face caused the perforation by blocking the gas fuel injection passage such that the fuel was forced around the back of the cap. Except for the perforation, no other area of the liner cap exhibited overly high temperatures based on metallurgical examination of the parts. Elimination of the fuel nozzle carbon buildup will prevent recurrence of this problem.

To some degree, all transition pieces had spalled regions of thermal barrier coating. The spalling is attributed to a greater

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than desired coating thickness, as determined by chip samples. The excessive thickness of the coating is detrimental to the adhesion of the coating. Until recently, the coating was applied to the transition piece with a hand operation, resulting in poor control of the coating thickness. To improve the process control of this operation, a robotic operation was developed, producing a more uniform, correct thickness coating. Subsequent visual inspections made in December 1991 showed the coating to be in excellent condition.

Closing

As stated, this was the most detailed engineering test of a new gas turbine. Combustion component evaluation was a major portion of this field test. Test results verified the results from individual component and single chamber development testing. These results confirm the basic soundness of the mechanical design and the ability of the combustor to meet emissions requirements. Finally, the causes underlying the areas of minor hardware distress seen at the first combustion inspection are understood and changes to address these findings will be implemented at the next scheduled machine outage.

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